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Hierarchical Power Co-Optimization and Management for LLM Chiplet Designs

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ABSTRACT

The demand for efficient and high-performance hardware for large language models (LLMs) has driven the development of scalable chiplet design, which requires careful power optimization and management. This paper presents a co-optimization and management methodology for hierarchical power delivery of chiplet designs targeting LLM applications. To model LLM workload mapping and power delivery, we first build a scalable chiplet simulator, which demonstrates different power strategies have notable efficiency impact and require careful and thorough optimizations. We further develop a co-optimization framework *ScalePoM* for chiplet power management. Based on given LLM model and PPA requirements, *ScalePoM* can automatically explore the chiplet architecture and workload mapping for optimal hierarchical power delivery. Our co-optimization methodology is evaluated through two scaled LLM chiplets with different interconnect topologies, achieving an average of 45% and up to 62% energy saving for large language model inferences with various sparsity levels.

KEYWORDS

Power Management, Co-optimization Methodology, Chiplets, LLMs

1 INTRODUCTION

The ever-increasing demand for large language models (LLMs), such as GPT-4 and its successors, has led to significant demands for efficient and accelerated hardware performance. The integration techniques for large-scale computing systems, such as multi-chip modules (MCMs) or chiplets, become crucial to meet this demand. As shown in Fig. 1, the chiplet technique integrates multiple chip dies via advanced 2D/2.5D or 3D packaging to scale up the computational performance [1, 2]. However, with the integration of large amounts of chiplets, efficient power delivery and management become complex and critical to ensure an energy efficient operation of these integrated systems [3, 4].

Conventional power solutions typically rely on board-level power management integrated circuits (PMIC) to step down the supply voltage from the PCB board (5-12V) to the processor cores (0.8-1.2V). However, it cannot provide flexible and dynamic management for

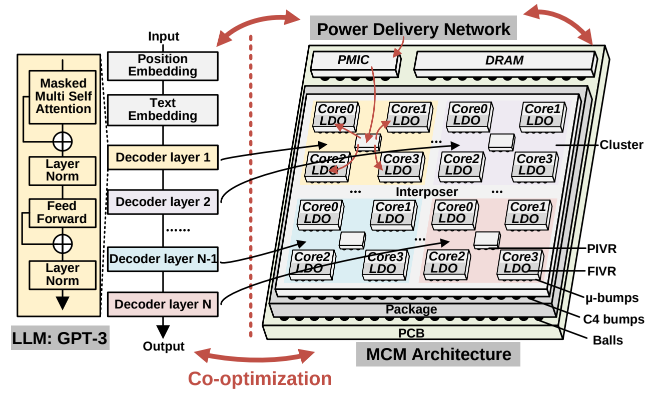

Figure 1: Chiplet modules and the deployment of LLMs.

specific requirements of each core. Commercial chiplet systems usually adopt two-level or three-level hierarchical power management. Beyond board-level PMIC, on-package integrated voltage regulator (PIVR) and on-chip fully integrated voltage regulator (FIVR) are often adopted, such as PIVR in Intel Ponte Vecchio [2] and on-chip LDO in AMD EPYC [1]. These hierarchical solutions form power delivery networks (PDNs) with independent and flexible voltage supplies for each chip, maximizing the efficiency of each chip and enhancing overall system performance.

Meanwhile, the emergence of large generative models, e.g., large language models (LLMs), proposes new requirements for efficient power management. Due to the significant operation number and weight storage, LLMs deployments, e.g. inferences, exhibit high latency and energy consumption. Especially, the LLMs often require token-by-token computation, resulting in a nonparallel workload and low utilization rate of the hardware resource. For instance, the decoder in OPT-175B can only have a 13% GPU compute utilization on an NVIDIA T4 (16GB) GPU [5]. Various approaches have been explored to enhance performance and efficiency, such as specialized hardware accelerators [6] or better model parallelism [5, 7].

To provide high-efficiency power delivery and management for complex integrated systems, such as LLMs computing modules, it becomes increasingly crucial to perform co-design/optimization to bring the power management and chiplet system design together considering the workload, energy efficiency, performance, and cost. Based on our analysis, it is observed that the power delivery and management solutions become non-trivial with a non-negligible overhead with increasing chiplet number. For example, the unbalanced workload mapping of LLMs may lead to the load current fluctuation and almost 10% efficiency degradation. Most prior work focuses on optimizing existing voltage regulation [4, 8], while none consider co-design/optimization for chiplet systems. The hierarchical power co-optimization and management need to be carefully

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ICCAD '24, October 27–31, 2024, New York, NY, USA

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ACM ISBN 979-8-4007-1077-3/24/10…$15.00
https://doi.org/10.1145/3676536.3676740

Table 1: Choices of regulators at each integration hierarchy.

	DLDO [11]	PIVR [2][12]	PMIC [13]
Input Vol. (V)	0.5-0.9	1.6-2.0	12
Output Vol. (V)	0.45-0.89	0.5-0.9	0.5-1.8
Load Curr. (A)	0.028-2.74	8-40	10-160
Peak Eff. (%)	89@2.74A	86@26A	91@160A
Resp. Time (µs)	1×10^{-3}	1	1×10^{3}
Area (mm^2)	0.126	14.7	112.5
Curr. Dens. (A/mm^2)	21.7	2.3	1.7

analyzed and selected based on trade-offs between efficiency and area to ensure reliable and efficient power distribution.

In this paper, we present a co-optimization methodology for hierarchical power delivery and management of chiplet designs targeting LLMs. First, we build a scalable chiplet simulator for LLMs considering different workload mapping and architecture choices. Based on this simulator, optimized power solutions can be explored based on our careful power delivery modeling, which incorporates two- and three-level PDN configurations. We also develop a co-optimization framework *ScalePoM* for power solution explorations to select the optimal chiplet and PDN topology with different workload mapping. Moreover, we utilize dynamic voltage and frequency scaling (DVFS) management to further enhance energy efficiency under performance requirements. Our co-optimization methodology is evaluated using two scaled Simba-like [9] and NN-Baton-like [10] chiplet modules, and is scalable to other chiplet architectures. The evaluations show our co-optimization framework can automatically explore the power solutions for LLM chiplets to obtain an average of 45% and up to 84.6% energy saving.

The contributions of this paper are summarized as follows:

- We build a chiplet simulator for LLMs considering workload mapping and careful power delivery modeling. Based on it, we can evaluate chiplet designs for efficiency, area, performance, dynamic droop, etc.
- We develop a co-optimization framework *ScalePoM* for chiplets to improve the efficiency of LLM computation combined with a chiplet simulator and power management explorer.
- We evaluate our methodology through two scaled chiplet designs for several LLMs. Online DVFS technology is also explored to further enhance energy efficiency.

2 BACKGROUND

Chiplet Architecture Solutions. Chiplet is a technique that leverages advanced package-level integration to scale up system performance effectively. Unlike a monolithic large die, a chiplets solution combines a number of smaller chiplets to form a large system, resulting in reduced fabrication and design expenses. Considering the rapid growth of AI model size, especially for LLMs, chiplet is a promising and important solution to provide scalable hardware performance. Several chiplet designs have been developed for DNN acceleration. For example, Nvidia Simba [9] is an MCM-based deep-learning inference chiplets, which consists of 36 chiplets for up to 128 TOPS at high energy efficiency. NN-Baton [10] is another chiplet solution with an automatic tool for DNN mapping on a Ring-based multichip accelerator with promising energy savings. However, prior chiplet designs only focus on conventional NN inferences, with the power ranges at 10W to 110W [9]. None of them explored chiplet optimizations for LLM applications, which could consume more than 300W for satisfied inference performance. At this power level, the design and selection of power delivery network and management strategy becomes particularly crucial.

LLM Mapping and Scheduling. LLMs typically employ autoregressive Transformer models, which have been widely used for text-to-image or text-to-video generations. The LLM inference involves serial token decoding, leading to performance and utilization challenges. Generating each token requires transferring all parameters from the storage to computation unit, leading to severe memory bandwidth bottleneck and low hardware utilization. To improve the LLM inference performance and throughout, numerous optimizations have been specifically explored. FlexGen enables high-throughput LLMs on a single GPU by adopting a new scheduling strategy to support a larger space of batch size choices [5]. Speculative decoding, by running the approximation models and the target large models in parallel, increases the parallelism of LLMs at each decoding step [14]. However, these complex mapping and scheduling strategies lead to imbalanced workloads, which actually requires careful considerations for power co-optimization and management. For instance, the workload mapping and scheduling status of each chiplets could be taken into design considerations to determine the power delivery topology and management solutions.

Power Management Solutions. Targeting a chiplet system with high power consumption, the power management solutions need to be carefully explored for efficient power delivery. A few percentage efficiency savings will bring substantial energy savings. Table 1 shows different regulator choices at each integration hierarchy. On-chip digital LDOs (DLDO) deliver load current for a few amps or less and have low area cost. However, it has low efficiency for a high voltage conversion ratio. Switching converters provide better efficiency at a large conversion ratio but only have a few fixed output voltages (capacitor-based) or require expensive discrete components (inductor-based). Rather than on-chip regulators, on-package PIVR can deliver tens of amps load current and has the capability of one-to-many management, i.e., group multiple chiplets as one voltage domain cluster.

Hierarchical power delivery and management strategy, as shown in Fig. 1, combines high voltage dropout PMICs and multiple power clusters regulated by DLDOs or PIVRs, offering high conversion efficiency and management flexibility. However, such hierarchical power solutions require systematic design explorations and co-optimizations. For instance, the imbalanced workload mapping might lead to significant voltage variations among chiplets in a unified voltage domain cluster with a single PIVR, undermining the system efficiency. Therefore, it is imperative to delve into cross-stack design and co-optimization tasks that encompasses model mapping and scheduling, chiplet architecture, and regulator circuits.

3 POWER MODELING AND ANALYSIS

3.1 Chiplet Architecture and Mapping for LLMs

To evaluate different power management choices, we first build a chiplet simulator targeting LLM applications with comparable performance with GPUs [5]. We refer to Simba [9] and NN-Baton

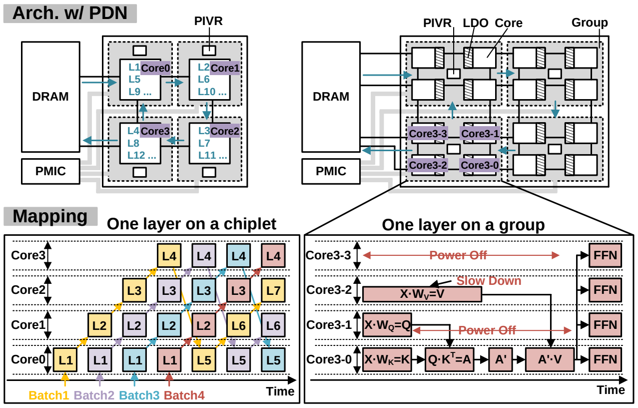

Figure 2: Chiplet design and the mapping for GPT-3 13B.

[10], which used representative mesh and ring interconnect topology, and build a larger scale chiplet architecture model for analysis and evaluation, as shown in Fig. 2. Our chiplet simulator is based on an in-house analytical performance model and performs simulations across package-, chiplet-, core levels, and a wide voltage operational range for chiplets. The chiplet design can be scaled for specific performance and area requirements, such as 3X and 12X larger chiplet designs than Simba [9] as listed in Table 2. Various micro-architecture parameters such as NoP bandwidth and performance are also expanded accordingly. For example, the network-on-package (NoP) bandwidth scales with the square root of the scale index, considering chip-to-chip ground-referenced signaling (GRS) transceivers are located at the top and bottom rows of each chiplet [9]. Meanwhile, the performance scales linearly with the scale index. The power consumption is also estimated and expanded based on the area and operating frequency, e.g., linear scale with the area index under the same operating frequency. For low voltage operations, we perform careful characterization of technology libraries using Cadence Virtuoso to obtain accurate voltage and frequency estimations. Referring to a GPU-level power budget, we first establish chiplet design cases with four large chiplets (57W) or 16 small chiplets (15.4W). Due to the limited space, we use the Simba-like chiplets for illustration in this section.

We first deploy a GPT-3 model [15] onto the built chiplet cases in our LLM chiplet simulator. Fig. 1 provides an overview of the GPT-3 model architecture, which comprises 96 layers of transformer decoders for GPT-3 175B and 40 layers for GPT-3 13B. Considering a 2×2 chiplet matrix, each chiplet has 48TOPS peak performance and is powered and managed by a PIVR. As shown in Fig. 2, we leverage a dataflow mechanism with ring topology to facilitate the GPT-3 13B inference. Employing a batch-level pipeline can also enhance hardware utilization and throughput [5]. As depicted in the computational graph at the lower left corner of Fig. 2, a batch size of 4 can maximize the utilization of available hardware resources in this specific application scenario. Adopting this per-layer mapping approach minimizes NoP communication between chiplets to reduce the data transfer overhead.

As for small chiplets, i.e., 4×4 chiplet matrix, we employ a partitioning scheme where each 2×2 array is designated as a power group with one PIVR. Each power group computes one layer of the

Table 2: Architecture parameters of our chiplet model.

Scale Index	Simba[9]	3X	12X
NoP Bandwidth (GB/s/Chiplet)	100	173	346
NoP Energy (pJ/bit)	0.82-1.75	0.82-1.75	0.82-1.75
Peak Perf./Chiplet (TOPS/Chiplet)	4	12	48
Core Power/Chiplet (W)	0.22-4.2	0.67-13	2.66-52
Trans. Power/Chiplet (W)	0.66-1.4	1.14-2.42	2.28-4.84
Area (mm^2)	6	18	72
# of power groups	-	4	4

GPT-3 13B model, equivalent to a large chiplet. Taking four chiplets group as an example, in the first time step, chips 0-2 perform parallel pre-attention linear layer calculations for query (Q), key (K), and value (V), i.e., KQV generation. The computational workload for each chiplet is nd^2, where n is the number of tokens per sentence and d is the length of the vector representation of each token. In the second and third time steps, attention layers are computed by chip 0, with each chiplet's computational workload being n^2d. Finally, in the fourth time step, all chiplets jointly execute a feed-forward network (FFN), with a total computational workload of $9nd^2$. In our experiment, GPT-3 13B model inference is set with a n of 32 and d of 5120; hence, the nd^2 and n^2d are around 839M and 5.2M.

3.2 Power Delivery Network for Chiplet

To evaluate the power efficiency of hierarchical power solutions, we implement a power delivery model for the commonly adopted 2.5-D integration technology for our chiplets, as shown in Fig. 3. Multiple accelerators and PIVR chiplets are integrated into an interposer and interconnected using NoP. The integrated chiplets are then packaged together and integrated with PMIC and DRAM onto a PCB, providing a complete, end-to-end solution for high-speed computing. Taking a three-level PDN as an example, the power from the PMIC is delivered to PIVRs through a series of interconnections, including the metal interconnects on the PCB, Balls, package substrate, C4 bumps between the substrate and interposer, and micro-bumps between the interposer and chiplet. Subsequently, the power from FIVRs is delivered through redistribution layers (RDLs) within the interposer to reach the on-chip FIVR, which further distributes power to various circuit modules by on-chip power meshes. We adopt a commonly used ladder RLC network model [8] to model the aforementioned physical structure using SPICE-level circuit simulation, as shown in Fig. 3. The included parasitic parameters are aligned with other PDN models [16] and calibrated to match the chiplet power.

The efficiency of power delivery is the most critical performance metric. The efficiency is determined as the proportion of power utilized by the load to the total power, as equations (1) and (2).

$$eff. = \frac{P_{load}}{P_{load} + P_{loss\cdot PMIC} + P_{loss\cdot PIVR} + P_{loss\cdot FIVR}} \tag{1}$$

$$P_{loss\cdot i} = P_{load\cdot i} \cdot \left(\frac{1}{eff_i} - 1\right) + Z_{grid\cdot i} \cdot I^2_{out\cdot i} \tag{2}$$

where P_{load} denotes the energy consumed by the load, while $P_{loss\cdot i}$ indicates the power dissipated in the delivery procedure for a given voltage stage, where i refers to PMIC, PIVR, or on-chip FIVR. The

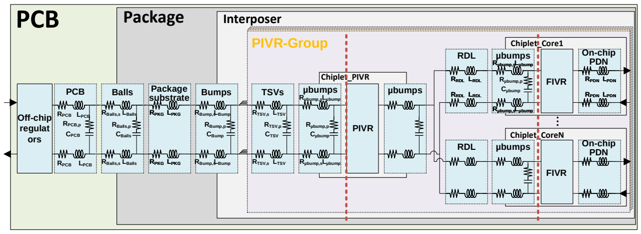

Figure 3: Power delivery modeling for chiplets with hierarchical power management.

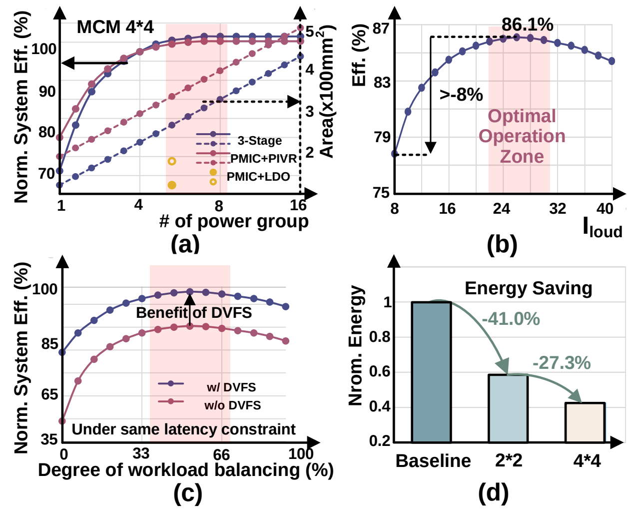

Figure 4: PDN modeling results and benefit assessments.

eff_i, $Z_{grid \cdot i}$, and $I_{out \cdot i}$ represent the efficiency of the regulator, the impedance of the transmission network, and output current, respectively, for their corresponding regulator. The regulator efficiency eff_i is based on the datasheet of off-shelf commercial regulators or published data, and the data of Z_{grid} is obtained by simulating the RLC network through SPICE simulations.

We also consider the area cost comprising the area of regulators and additional package overhead introduced by the PIVRs as equation (3). Each regulator's area is estimated by the product of current density and load current.

$$Area = A_{PMIC} + N_{PIVR} \cdot \left(A_{PIVR} + A_{Package}\right) + N_{FIVR} \cdot A_{FIVR} \tag{3}$$

In addition, we model the dynamic voltage droop triggered by transient current grows with current, as equation (4). Hence, when the chiplet switches from low to high load, or vice versa, the voltage droop can be tracked and addressed by dynamic power management. We observe the dynamic voltage and frequency scaling (DVFS) management could be co-optimized with model mapping to bring promising energy saving.

$$V_{droop} = \frac{Q}{C} = \frac{\int I_{tr} \mathrm{d}t}{C} \approx \frac{\bar{I_r} \cdot t_{resp}}{C} \tag{4}$$

3.3 Analysis and Observations

Based on our chiplet simulator and power delivery modeling, we explored several design trade-offs between system efficiency and power management solutions. Some observations are summarized as follows and in Fig. 4. This analysis is based on GPT-3 13B inference, while the observations hold true for other LLMs.

System efficiency is greatly impacted by power topology. As shown in Fig. 4(a), when the total count of chiplets is constant, allocating more chiplets into a single power group leads to fewer PIVRs, i.e., fewer power groups. Although this reduces the allocated area for power management, this topology also leads to efficiency degradation due to the following two issues. The first issue is higher impedance of power network due to the increased distance between PIVR and chiplets, which leads to larger energy loss along power delivery path. The second issue is larger static IR drop and dynamic voltage droop V_{droop} (4) due to the sense error and feedback loop delay of the PIVR for far distance loads, which necessitate an increased voltage margin and reduced energy efficiency [17]. The system efficiency can be improved with an increase in power groups and eventually stabilizes. Beyond this certain point, adding more power groups only results in increased area without further efficiency gains. When considering power delivery for the chiplet with the same load current, using PMIC and PIVR exhibits high efficiency in most cases due to the absence of energy loss from FIVRs. However, this combination also incurs high area costs due to PIVR exhibiting a lower current density than FIVR.

Workload mapping also greatly impacts system efficiency. Fig. 4(b) depicts that the efficiency of a single buck converter is greatly varied with the loading current. The optimal efficiency operating zone is narrow and only under a specific load current. The power efficiency quickly declines (potentially exceeding 8%) at currents below or above this zone. Hence mapping the workload with operating at optimal efficiency region is desired, as explored in Fig. 4(c). The term "degree of workload balancing" in Fig. 4(c) refers to the distribution degree of running workloads across multiple chiplets. For instance, a more balanced workload mapping implies a greater number of chiplets concurrently processing a single layer of an LLM. Taking Fig. 2 as an example, a workload balance of 0% corresponds to a single chiplet, 25% to four chiplets, and 100% to sixteen chiplets, all running a single layer. It is observed that the optimal efficiency is narrow and closely related to the mapping strategy. For instance, when mapping a single layer to a portion of chiplets, the inactive chiplets will stay idle and degrade the system efficiency. Conversely, when mapping a single layer to a large number of chiplets, the increased latency due to data communication and NoP energy consumption also result in an efficiency degradation. Given the unbalanced workload mapping, DVFS is a helpful technology to enhance efficiency by dynamically adjusts the voltage and frequency according to the operational state of the chiplets, reducing the energy consumption of inactive chiplets.

Chiplet power solutions need to be co-optimized with mapping and chiplets architecture. Since the power efficiency is impacted by so many factors, the power solution for chiplet system requires systematic co-design and optimizations together with hardware architecture and workload mapping. For instance, the four large chiplets in Fig. 2 are inclined to use a two-level power structure with off-chip PMIC and PIVR to achieve an efficiency of 76.8%. On the other hand, the 16 smaller chiplets prefer a three-level

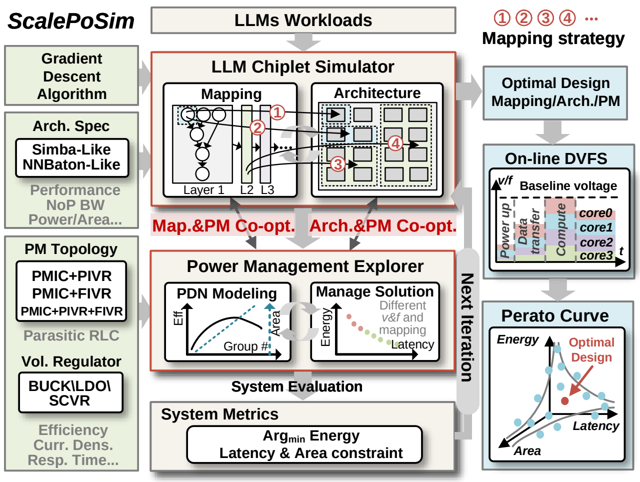

Figure 5: Overview of the co-optimization framework for chiplet power management.

power delivery with off-chip PMIC, PIVR, and on-chip FIVR with finer-grained DVFS to obtain the peak efficiency of 75.3%. Compared with a baseline design with single-level PMIC, Fig. 4(d) shows that the two-level management can achieve 41.0% energy reduction, and the three-level solution further obtains 27.3% reduction.

4 CO-OPTIMIZATION FRAMEWORK FOR CHIPLET POWER SOLUTIONS

4.1 Power Co-Optimization Framework

The above analysis indicates that systematic co-design and optimization are required for efficient power delivery of chiplet systems. In this section, we present our co-design/optimization framework *ScalePoM* for chiplet power solutions, as shown in Fig. 5. The framework leverages our LLM chiplet simulator and PDN modeling to automatically search the optimized mapping strategy, chiplet architecture, and PDN efficiency. Our chiplet simulator (explained in Section 3.1) takes LLM workloads as input and performs evaluations for different mapping strategies and chiplet architectures. Combined with a power management explorer, which includes PDN modeling and management strategies, *ScalePoM* explores various power delivery and management solutions and generates system metric results. Due to the large search space involved, exhaustive search is not practical. Therefore, we adopt a gradient descent algorithm for design space exploration (DSE).

The evaluation of *ScalePoM* mainly accommodates two chiplet architecture topologies: mesh-based and ring-based. Three types voltage regulators (BUCK, SCVR, and LDO) and a range of hierarchical power management topologies can be selected for power solution evaluations. The inputs for *ScalePoM* consist of architectural specifications, including performance, NoP bandwidth, power, and area for a single chip, as well as PDN parameters like parasitic resistance, inductance, and capacitance. Additionally, it considers voltage regulator parameters such as efficiency, current density, and response time. We also incorporate an online DVFS scheme that adjusts the operating voltage and frequency of different chiplets

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Algorithm 1 Power co-optimization and management for chiplets

Require: **LLM workload parameter** $[D_{model}, \#L, \#H]$;
  Power budget P_{budget};
  Architecture range $[2\times2,\ 3\times3,\ 4\times4,\ 5\times5,\ 6\times6, ...]$;
  NoP topology $[mesh,\ ring]$;
  PDN topology $[PMIC+PIVR, PMIC+FIVR, PMIC+PIVR+FIVR]$;
  Latency & Power & Area constraint;
  Voltage v; **Frequency** $freq$.
Ensure: **Optimal design** $\boldsymbol{Design_{opt}}$

1: Architecture index $m=0$; Mapping strategy index $n=0$; PM index $p=0$
2: **for** $m: 0 \rightarrow len(Archi\_Range)$ **do**
3:   Select an architecture $Archi_m$ ▷ **Step 1.**
4:   Calculate the Perf., Power, NoP BW, and Area of each chiplet based on num_{chip}
5:   Calculate eff. v.s. num_{pg}: $Eff.\left(num_{pg}\right)$ ▷ **Step 2.**
6:   $Eff.\left(max\right) \rightarrow num_{pg}$
7:   Calculate the mapping range: $Mapping\_Range = (\alpha + (\#H-1)) \times num_{chip} \times num_{batch}$ ▷ **Step 3.**
8:   **for** $n: 0 \rightarrow len(Mapping\_Range)$ **do**
9:     Select a mapping strategy $Mapping_n$
10:     **Algorithm 2** $\rightarrow PDN_{Curr}, FoM_{Curr}$ ▷ **Step 4.**
11:     **if** $FoM_{Curr} > FoM_{best}$ **then** ▷ **Step 5.**
12:       $FoM_{best} = FoM_{Curr}$; $PDN_{best} = PDN_{Curr}$;
13:       $Design_{opt} = \left\{Archi_m, Mapping_n, PDN_{best}\right\}$;
14:     **end if**
15:     Calculate the gradient of the FoM
16:     Give direction for Mapping & PM co-opt. of next iter.
17:     **if** Not improve after N iterations **then**
18:       **break**;
19:     **end if**
20:   **end for** ▷ **Step** 3 $\rightarrow$ 5.
21: **end for** ▷ **Step** 1 $\rightarrow$ 5.
22: **return** $Design_{opt}$

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depending on the workload, further enhancing the runtime system efficiency. Finally, after several iterations, *ScalePoM* is able to obtain an optimal Pareto design set for the target LLMs.

Algorithm 1 outlines the end-to-end power co-optimization flow of our framework. In the beginning, the initial architecture definition of chiplets is selected from the design space, as Step 1 is shown in line 3. With the power budget constraint, the performance, power, NoP bandwidth, and area of each chiplet are explored based on specified chiplet number (num_{pg}). The PDN modeling and optimal number of power groups in Step 2 are initially determined based on the relationship between system efficiency, as illustrated in Fig. 4(c). At this step, the choice of power management primarily relies on the impedance evaluations of the PDN, i.e., from line 5 to line 6. The impact of model mapping strategies is evaluated from line 7 to line 9 as Step 3. Workloads with varying computational requirements are assigned to hardware at different granularity. Initially, we estimate the mapping space considering chiplet number (num_{chip}), LLM model layers ($\#H$), batch size (num_{batch}), and specific mapping strategies (α), where α denotes the division number for a single layer segmentation. For instance, the α value for the

Algorithm 2 PDN modeling and management solution

Require: mapping strategy map, $I_{load \cdot chip}$, N_{core};
activity α; **power on ratio** γ;
Average voltage v;
$eff_{PMIC}(I)$, $area_{PMIC}$, $eff_{PIVR}(I)$, $area_{PIVR}$, $eff_{FIVR}(I)$, $area_{FIVR}$

Ensure: Best Power delivery network PDN_{best}, FoM_{best}

1: iteration number $i = 0$; $FoM_{best} = 0$; $PDN_{best} = Default$

/* *Compute min & max granularity of power domain (PDG)* */

2: $\min(PDG) = \max(core, map(operator))$

3: $\max(PDG) = \max(core, map(layer))$

4: Number of PMIC: $N_{PMIC} = \left\lceil \frac{I_{load}}{I_{PMIC \cdot output}} \right\rceil$

5: Number of PIVR: $N_{PIVR} = \left\{0, \left[\frac{N_{core}}{\min(PDG)}, \frac{N_{core}}{\max(PDG)}\right]\right\}$;

6: Number of FIVR: $N_{FIVR} = \{0, N_{core}\}$

7: $PDN\_Spc = [N_{PMIC} \cdot PMIC + N_{PIVR} \cdot PIVR + N_{FIVR} \cdot FIVR]$

/* *Calculate the eff./area of PDN for typical scenarios* */

8: **for** $i : 0 \rightarrow len(PDN\_Spc)$ **do**

9: $\quad PDN = PDN_i$;

10: $\quad eff_i = eff(I_{load \cdot chip}, Z_{grid})$

11: $\quad area_i = area(N_{PMIC}, N_{PIVR}, N_{PIVR})$

12: /* *Considering DVFS* */

13: $\quad energy_i = \alpha \cdot v^2 / eff_i \cdot \gamma$;

14: $\quad FoM_{Curr} = 1/(energy_i \cdot area_i)$

15: $\quad$ **if** $FoM_{Curr} > FoM_{best}$ && $constraint_{meet}$ **then**

16: $\qquad FoM_{best} = FoM_{Curr}$;

17: $\qquad PDN_{best} = PDN$

18: $\quad$ **end if**

19: $\quad$ **return** PDN_{best}, FoM_{best}

mapping strategies in Fig. 2 is set to 4, reflecting the parallel processing of four tasks. Additionally, the batch size should be maximized as much as possible to improve hardware utilization while satisfying the latency constraints. Step 4 involves the optimization of the power distribution network and management solutions, with details in Algorithm 2. The optimal PDN is chosen based on the current architecture and mapping strategy, with a primary focus on the impact of load current on system efficiency. Step 5 encompasses the evaluation of system metrics, focusing on identifying the optimal design that maximizes the Figure of Merit (FoM). At the end of each iteration, the gradient of the FoM is calculated to guide the direction for the subsequent co-optimizations, which includes adjusting the workload allocation on a single chip and batch size. The above steps are iteratively performed to generate a Pareto design set to achieve minimum energy consumption.

Algorithm 2 provides the exploration details for PDN and management solutions, i.e. Step 4. The workload mapping and architecture are fed into the power management explorer to evaluate and explore various power management options. As shown in line 3 to 8, we determine the factors for the minimum and maximum granularity of the power domain based on mapping and use them to explore the number intervals of PMIC, PIVR, and FIVR. Different PDN delivery options including two-level and three-level architectures as well as different grouping strategies are traversed during exploration for efficiency and area, as shown from line 10 to line 21. During the analysis, a system FoM metric is used for evaluation which contains both system energy and area, and the maximum FoM is eventually identified. The target FoM can be chosen as either energy-driven or energy-area-product-driven.

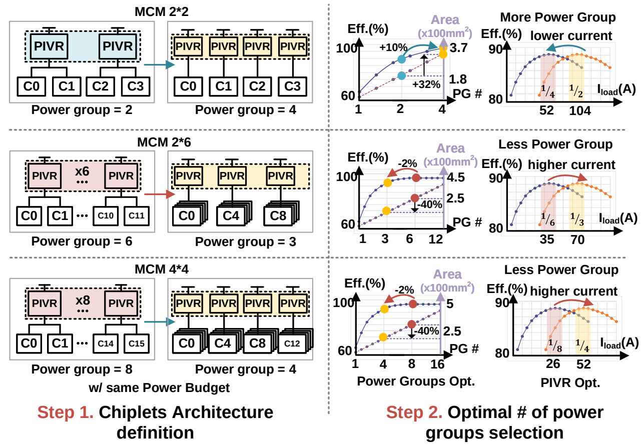


Figure 6: Co-optimizations between the chiplet power and architectures.

4.2 Co-optimize Chiplet Power with Mapping and Architecture

The co-optimization Steps in Algorithm 1 are further explained by the following experiments and Fig. 6-Fig. 7.

Chiplet power and architecture co-optimization. Following the chiplet architecture definition, the PDN modeling and selection of power groups is initially optimized for maximum efficiency. Fig. 6 showcases three distinct MCM topologies: 2×2, 2×6, and 4×4 chiplets, operating with the same power budget. The performance and power consumption of chiplets scales proportionally as simulated by our LLM chiplet simulator. In this experiment, one PIVR is allocated to power two chiplets, serving as the baseline design. It is apparent that the optimal power solution is influenced by the number of power groups across various architectures. For instance, in the case of a 2×2 chiplets, each chiplet has relative large area, leading to increased parasitics in PDN and reduced efficiency. Constructing the system with 4 power groups can reduce the load current for individual chiplet and enhance efficiency by approximately 10%, albeit at the cost of a 32% increase in power management area cost. For 2×6, and 4×4 chiplet matrixs, the optimal power group designs are explored as 3 and 4, respectively. Compared to the baseline, the explored power topology results in a slightly efficiency loss of 2%, while achieves a significant area savings of approximately 40%. Meanwhile, regulators require adjustments to ensure operating within the optimal efficiency zone across varying current demands.

Chiplet power and mapping co-optimization. Once the architecture and the respective number of power groups are determined, LLMs workloads with varying computational requirements are distributed across hardware with different mapping granularity, as shown in Fig. 7. At this step, the specific operation states of each chiplet could be analyzed, e.g. whether the chiplets are more

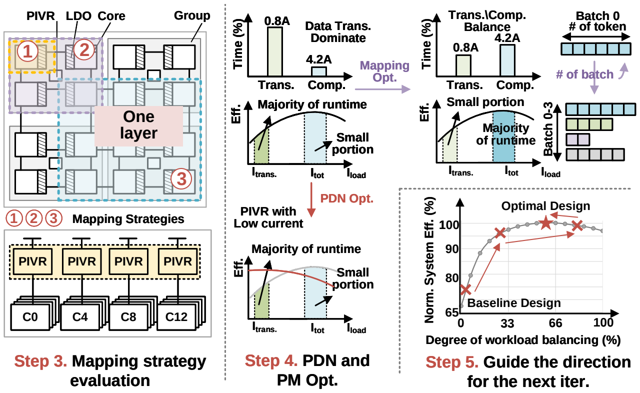


Figure 7: Co-optimizations between the chiplet power and mapping strategies.

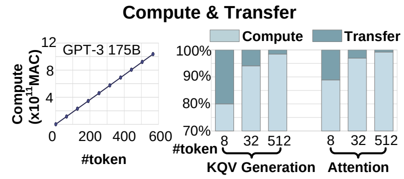
engaged in data transmission or computation task during specific time period. Due to different power consumption between data transmission and computation, as shown in Table 2, corresponding current distribution varies. For example, the KQV generation layer often has a higher *trans./comp.* ratio, in which the majority runtime is dedicated to data transmission between chiplets. Under such operation status, the optimal efficiency zone for the regulator differs from the average current assessed in Step 2. Therefore, PIVRs with lower current can be leveraged to enhance efficiency in Step 4. Furthermore, to enhance hardware utilization and reduce weight loading cost, batch-level parallel computation can also be employed to improve hardware utilization. Compared to a batch size of 1, simultaneously 4 batches computing increases the computation workload, matching the workload better with the regulator designs. The above mapping co-optimizations are all conducted to ensure the regulator operates within its optimal efficiency zone. Finally, Step 5 evaluates the current design metrics and provides guidance for the subsequent co-optimization iteration. Gradient descent algorithm is used to identify the optimal efficiency point without the need to explore the entire mapping space.

5 EVALUATIONS

5.1 Experiment Setup

Workloads: To evaluate the effective benefits of our methodology, we applied our *ScalePoM* co-optimization framework for three LLMs, i.e., GPT-3 175B [15], LLaMA [18], and T5 [19]. As shown in Fig. 8, GPT-3 and LLaMA are decoder-only architectures, and T5 is an encoder-decoder LLM in which encoder layers are incorporated before the decoder to enhance semantic understanding capabilities. Additionally, it can be observed computation numbers of GPT-3 vary significantly with increasing token numbers. Furthermore, the ratio between computation and data movement for different operators changes as the token numbers fluctuate.

Design Exploration Space: Our chiplet simulator evaluates a wide range of chiplet architectures and mapping strategy options, in which the mapping strategy supports different pipeline stages and batch sizes to enhance parallelism and hardware utilization. The hierarchical power management combining high voltage dropout PMIC and multiple power clusters with per-chip voltage tuning



LLMs Parameters

	Size	D_{model}	#L	#H	Category
GPT-3 [14]	175B	96	12288	96	decoder
LLaMA [16]	65B	80	8192	64	decoder
T5 [17]	11B	24	1024	128	encoder-decoder

Design Exploration Space

MCM	2x2, 2x3, 2x4, 3x3, 4x4, ⋯ (16nm)
Mapping	Pipeline Stage Batch Size Mapping granularity
PDN	2-stage \ 3-stage Buck \ SCVR \ LDO (16nm)
Voltage	0.41~1.2V
Frequency	161~2001MHz
NOP	Mesh\Ring
DRAM	8.75 *pJ/bit* [10]

Figure 8: Evaluation models and design exploration space.

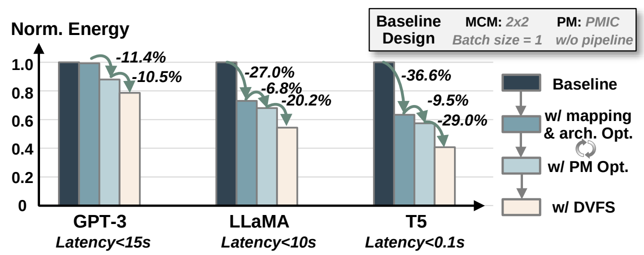


Figure 9: Co-optimization benefits for three LLM models.

provides the highest flexibility to pursue high system efficiency. Additionally, our power co-optimization framework supports low-voltage DVFS explorations. The data for chiplet cores and on-chip FIVR is primarily based on TSMC 16nm, and PMIC and PIVR data are collected from off-shelf product datasheets as in Table 1.

5.2 Co-optimization Benefits for Chiplets

Fig. 9 illustrates the power reduction benefits through different co-optimizations for three LLMs with different latency requirements. A 2×2 LLM chiplets architecture with off-chip PMIC and batch size of 1 is set as the baseline. Overall, 21.3%, 45.7%, and 59.3% energy savings are obtained, respectively, for different LLMs through co-optimization steps. We observe that each optimization step obtains varying benefits for different models and latency constraints. For instance, PM optimization and DVFS dominate the benefits achieved for the GPT-3 model. However, for LLaMA and T5, the optimization of mapping and architecture results in notable benefits. Even with a fixed chiplet configuration and mapping strategy, our co-optimizations for PM and DVFS can still bring energy savings from 20% to 38%.

Table 3 shows the details of optimized chiplet designs based on our framework for GPT-3, targeting a Simba-like architecture. It can be observed that the explored optimal design exhibits different design choices under different optimization objectives. For instance, four large chiplets are selected for the scenarios with tight latency constraints due to the lower communication overhead. A batch size of 1 is employed to minimize the inference latency. The power management scheme of PMIC with PIVR is deemed suitable for achieving effective inter-chip isolation and DVFS. In contrast, with relaxed latency, sixteen small chiplets are a better choice for 3-stage power management to exploit finer-grained DVFS to improve

Table 3: Optimal design for GPT-3 (# of tokens = 32).

Spec	Latency<15s	Latency<30s	Area·*Energy*
MCM	2×2	4×4	4×4
PM	PMIC+8PIVR	PMIC+8PIVR+16DLDO	PMIC+16DLDO
# of Power Groups/PD	4/4	4/16	16/16
PDN Eff.(%)	76.7	75.3	62.3
Voltage (V)	0.8-0.9	0.6-0.9	0.6-0.9
PM Area (mm^2)	303	312.8	132
Total Latency (s)	13.9	29.8	29.8
Batch Size	1	4	4
Energy saving(%)	23.3	38.4	34.8

Table 4: Optimized design for different LLMs with Simba-like and NN-Baton-like chiplet design.

Archi.	Simba-like		NN-Baton-like	
model	LLaMA	T5	LLaMA	T5
MCM	2×2	4×4	2×4	2×6
PM	PMIC+ 8PIVR	PMIC+8PIVR +16DLDO	PMIC+ 8PIVR	PMIC+ 8PIVR
Total Latency (s)	8.6	0.06	9.9	0.08
Throughput (token/s)	0.23	66.7	0.20	50
Batch Size	2	4	2	4
Energy saving(%)	45.7	59.3	38.3	66.0

efficiency. A batch size of 4 is set to enhance parameter utilization while incurring only a slight latency penalty. Furthermore, when the design target shifts to $area * energy$, PIVR is removed from the PM scheme, resulting in a 57.8% area gain (under 30s constraint).

We also evaluate our methodology for different chiplet topologies. Both mesh based Simba[9]-like and ring based NN-Baton[10]-like architectures are performed the power co-optimizations based on our framework, as the results shown in Table 4. Due to the decreased flexibility of ring-based architectures, NN-Baton-like architecture often incur a latency penalty. Additionally, due to the relatively smaller network scale of T5 and the convenience of accommodating around four layers of parameters on the entire MCM, it can fully leverage the parameters and achieve high efficiency.

5.3 Dynamic Power Management Strategies

To evaluate the dynamic power co-optimizations, we further evaluate voltage droop using SPICE simulation based on the power delivery model. Fig. 10 shows the voltage map at a certain moment during run-time for T5 inference. Before optimization, all 16 chiplets were powered by off-chip PMIC, indicating that an unbalanced load would cause significant droop in the computing cores and induce droop in the neighboring cores. After co-optimizing PM and mapping, PIVRs form different power groups to achieve relatively balanced workloads, while the load imbalance between groups is isolated by PIVR for less energy loss.

Additionally, DVFS is introduced to lower the voltage of low-load groups. We observe great potential for DVFS applications due to the non-parallel working mode of token-by-token processing. In the scenario of a large chiplet architecture where a single layer is mapped onto a chiplet, it is noteworthy that all chiplets could concurrently execute different layers for different batches, respectively. When the number of tokens varies between different batches, the

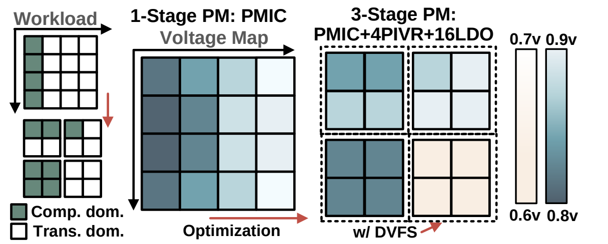


Figure 10: PM and mapping co-optimization for T5 and the voltage map at a certain moment during runtime.

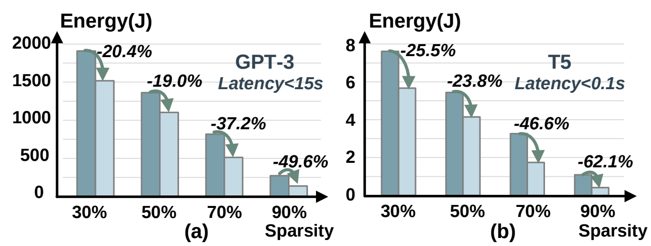


Figure 11: Energy saving across different sparsity levels.

number of operations to be computed and the amount of data to be transferred between chiplets also differ. Given that the bottleneck in a pipeline operation mode is determined by the slowest segment, it becomes feasible to adaptively reduce the frequency and voltage for chiplets burdened with fewer tokens while conversely elevating the frequency and voltage for the chiplet that is processing the greatest number of tokens.

We also evaluated our method across different sparsity levels and used the optimized design obtained in Fig. 9 as the baseline. Here, we present the evaluation results for GPT-3 and T5 models only, as the results obtained for LLaMA were comparable to GPT-3. Our co-optimization ensures that voltage regulators at each stage operate with maximum efficiency under different sparsity levels. Combined with DVFS, GPT-3 and T5 models can achieve up to 49.6% and 62.1% improvements when the sparsity reaches 90%.

6 CONCLUSION

In this paper, we present a co-optimization methodology for hierarchical power management of chiplet designs targeting LLM applications. A scalable chiplet simulator is built for LLMs considering workload mapping and careful power delivery modeling. We also develop a co-optimization framework *ScalePoM* for power delivery and management exploration to select the optimal architectures and PDN topology with an optimal workload mapping. Our method is evaluated through two scaled LLM chiplet accelerators with different interconnect topologies. The evaluation shows a significant 45% average energy savings can be obtained with the help of our co-optimization methodology.

ACKNOWLEDGEMENT

This work was supported in part by National Key R&D Program 2022YFB4400600, NSFC Grant No. U23A6007, 92164301, 62225401.

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